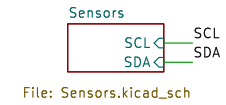
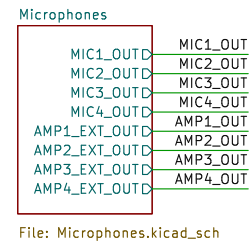
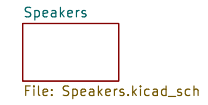
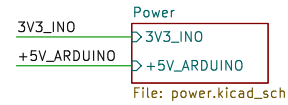
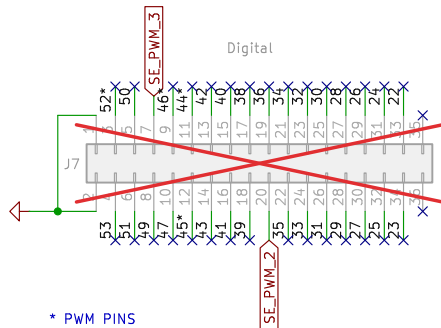
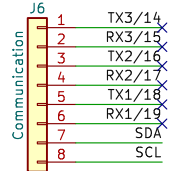
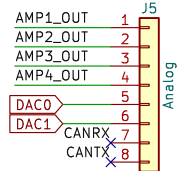
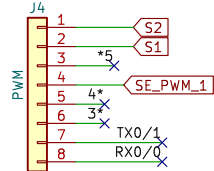
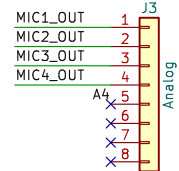
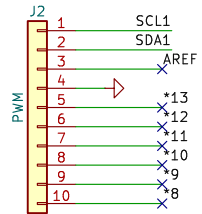
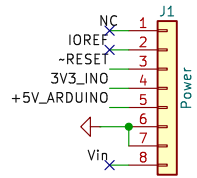


ARDUINO CONNECTORS



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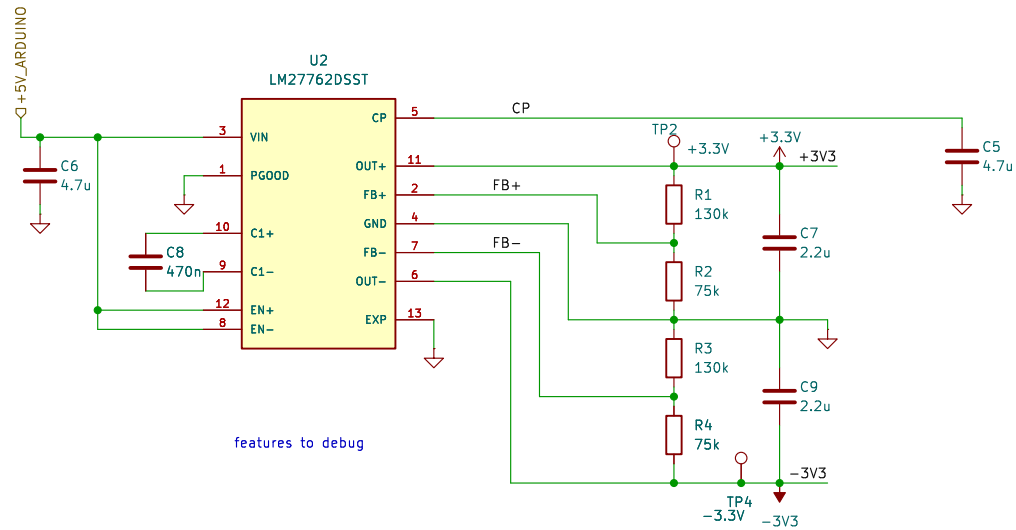
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Title: SensEdu

Size: A4 Date: 2026-02-26
 KiCad E.D.A. 9.0.5

Rev: 0.9
 Id: 1/6

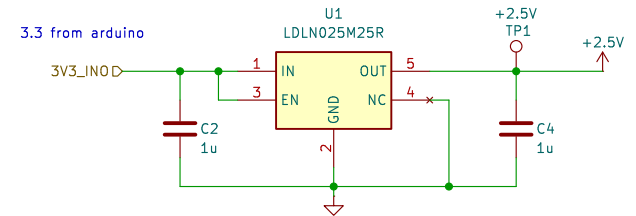
CHARGE PUMP + LDO FOR GENERATING $\pm 3.3V$



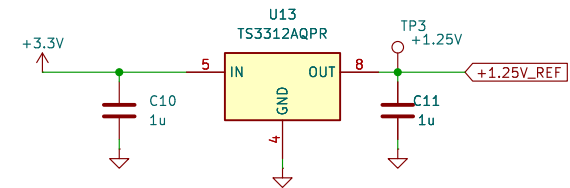
$$V_{out-} = -1.2V * (R39+R40)/R40; R40 > 50k\Omega; R40 = 75k\Omega; R39 = 130k\Omega \text{ OR } R40 = 91k\Omega; R39 = 160k\Omega$$

$$V_{out+} = 1.2V * (R41+R42)/R42; R42 > 50k\Omega; R42 = 75k\Omega; R41 = 130k\Omega \text{ OR } R42 = 91k\Omega; R41 = 160k\Omega$$

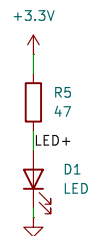
LOW DROPOUT VOLTAGE REFERENCE 3.3V $\rightarrow$ 2.5V



LOW DROPOUT VOLTAGE REFERENCE 3.3V $\rightarrow$ 1.25V



POWER INDICATION



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Sheet: /Power/
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Title: SensEdu

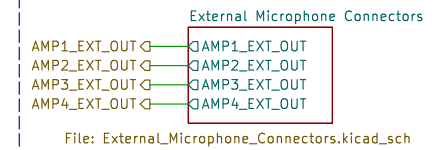
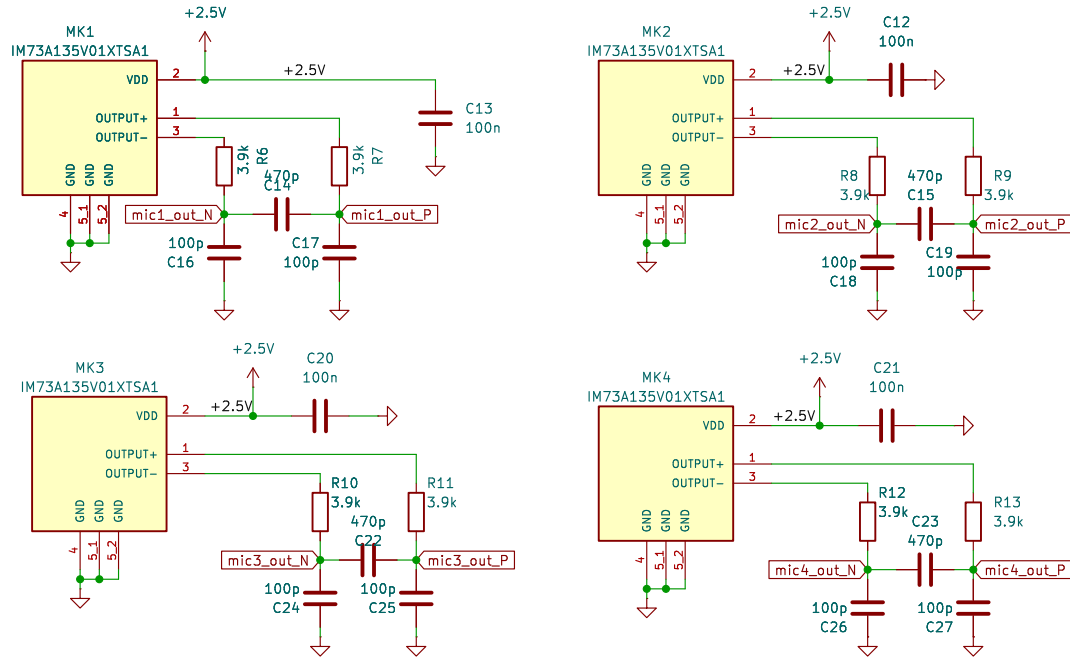
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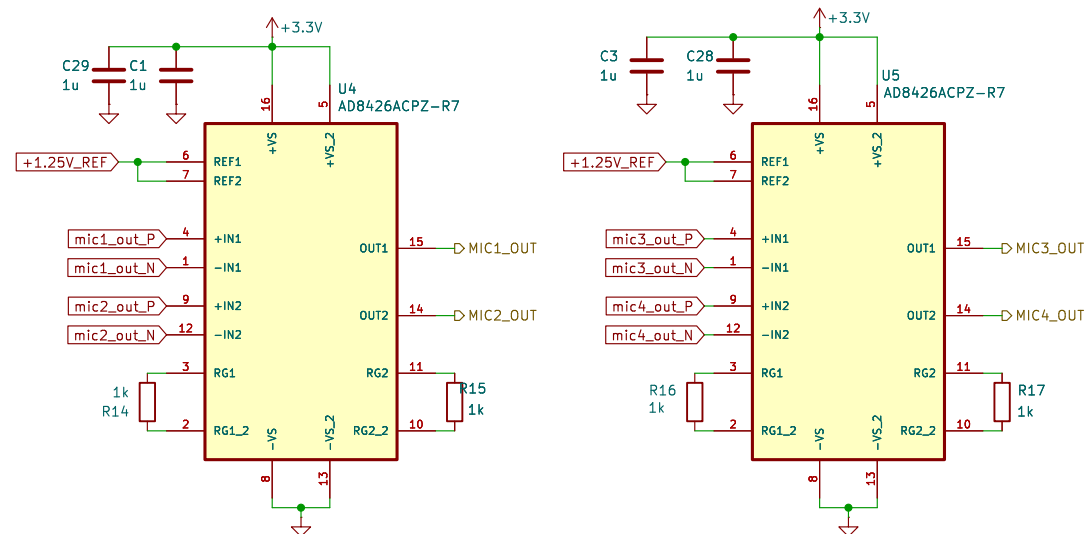
Rev: 0.9

Id: 2/6

MEMS MICROPHONES



MICROPHONE AMPLIFIERS



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Sheet: /Microphones/
 File: Microphones.kicad_sch

Title: SensEdu

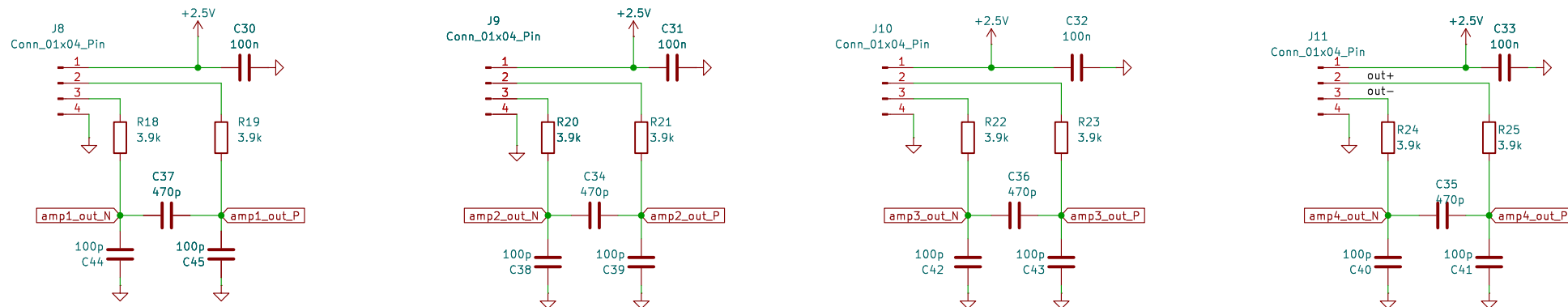
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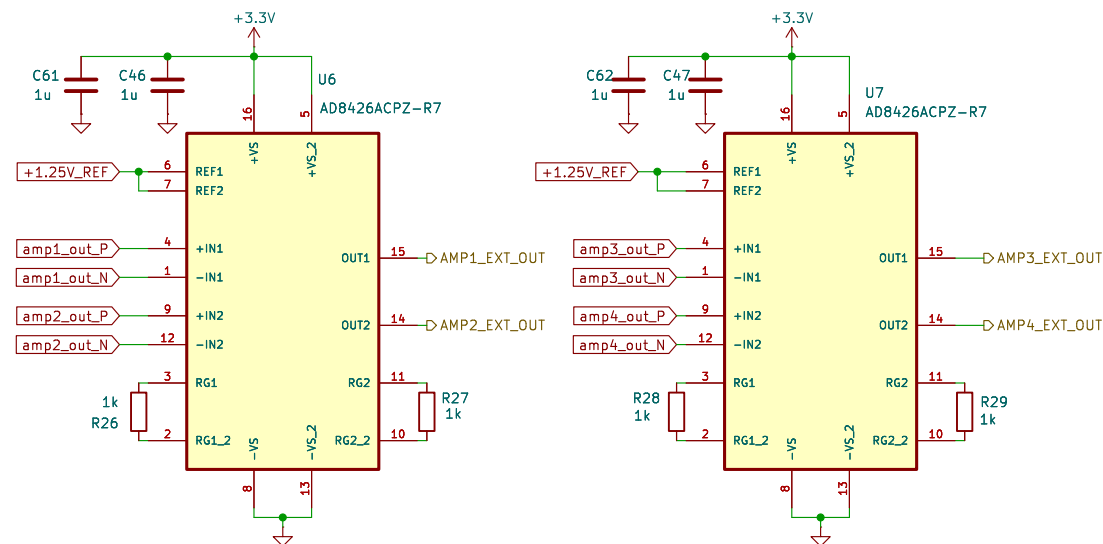
Rev: 0.9

Id: 3/6

EXTERNAL MICROPHONE CONNECTORS



MICROPHONE AMPLIFIERS



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Sheet: /Microphones/External Microphone Connectors/
 File: External_Microphone_Connectors.kicad_sch

Title: SensEdu

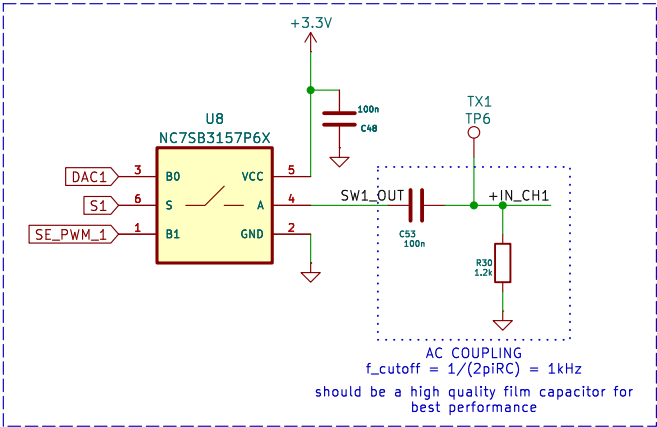
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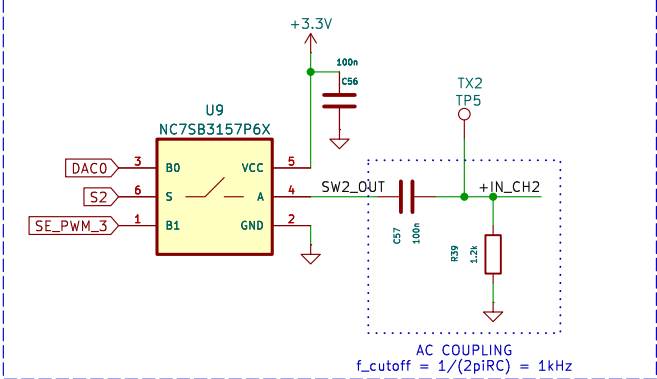
Rev: 0.9

Id: 4/6

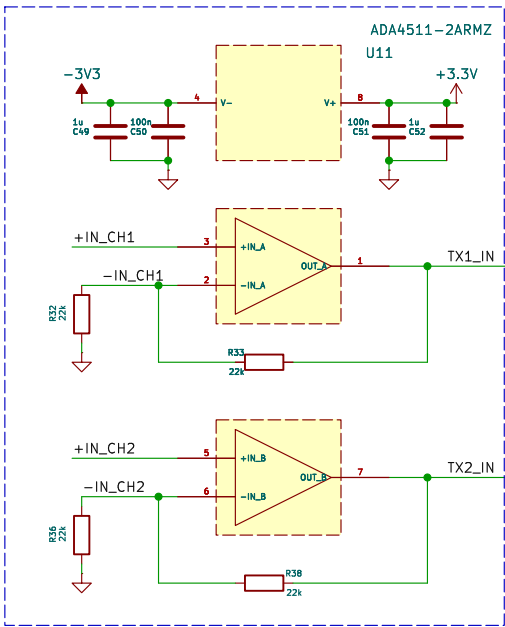
SWITCH WITH FILTER FOR DIFFERENT DRIVING SOURCES



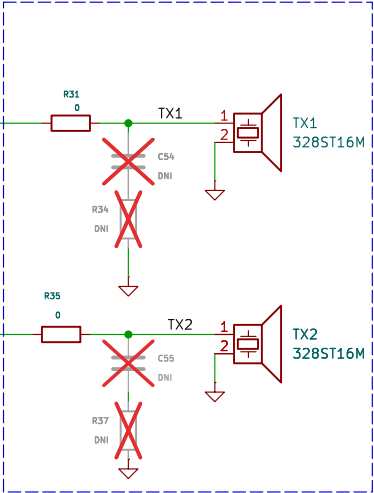
SWITCH WITH FILTER FOR DIFFERENT DRIVING SOURCES



DUAL CHANNEL PREAMPLIFIER



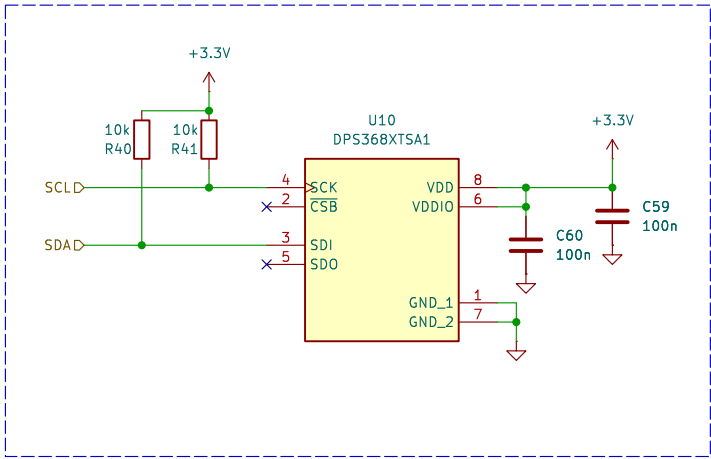
SPEAKERS



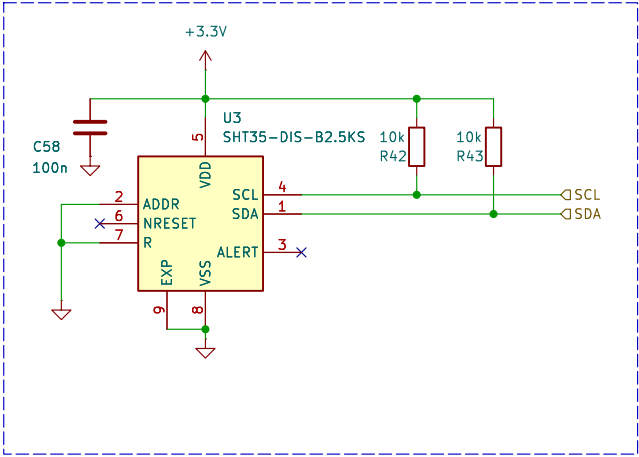
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Title: SensEdu	
Size: A4	Date: 2026-02-26
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PRESSURE SENSOR



HUMIDITY SENSOR



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